

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	27.08.2026	Duration:	60 minutes

Code of Conduct

I V.Rakshana(192425137) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Rakshana

Problem Statement 24: Smart Museum Artifact Preservation Platform

1. Introduction

The Smart Museum Artifact Preservation Platform is a digital system for registering, organizing, searching, monitoring and preserving museum artifact information. It can manage artifact metadata, images and documents, preservation condition, conservation activities, environmental observations, user roles, reports and audit history. Because museum records may have cultural and historical value, the system must be functionally correct, secure, reliable and easy to use.

This document defines a comprehensive test case design covering normal, boundary, invalid and exceptional conditions. It also applies standard black-box test-design techniques and provides requirement traceability and coverage.

2. Testing Objectives

- Verify all major functional requirements.
- Verify security and role-based access.
- Detect invalid input and boundary defects.
- Verify data integrity and reliable error handling.
- Evaluate performance, usability, compatibility, availability and recovery.
- Provide traceable coverage from requirements to test cases.

3. Functional Requirements

ID	Requirement	Test Focus
FR-01	Registration/Login/Logout	Credentials, validation, sessions
FR-02	Role-based access	Admin, curator/conservator, viewer permissions
FR-03	Artifact registration	Required fields, unique ID, validation
FR-04	Artifact update	Editing and persistence
FR-05	Archive/Delete	Authorization, confirmation, archive state
FR-06	Digital file management	Upload, type/size, retrieval
FR-07	Search and filtering	Keyword, category, condition, period, combined filters
FR-08	Artifact details	Metadata, files, condition and history
FR-09	Preservation status	Valid/invalid state transitions
FR-10	Conservation records	Create, view and update treatment history
FR-11	Environmental observations	Temperature/humidity validation
FR-12	Reports/dashboard	Filters, totals, charts and export
FR-13	Alerts/notifications	Threshold alerts and acknowledgement
FR-14	Audit trail	User, action, time and target record
FR-15	Database persistence	CRUD consistency and transactions
FR-16	Error handling	Safe messages and no corruption

4. Non-Functional Requirements

ID	Category	Test Objective
NFR-01	Performance	Search/view response meets defined target
NFR-02	Scalability	Stable behavior as records/users/files grow
NFR-03	Security	Authentication, authorization, input and file security
NFR-04	Reliability	No corruption during failures
NFR-05	Availability	Service meets defined availability target
NFR-06	Usability	Clear navigation, forms and messages
NFR-07	Compatibility	Supported browsers/devices work consistently
NFR-08	Maintainability	Useful logs and traceable failures
NFR-09	Backup/Recovery	Data can be restored accurately

5. Test Scenarios

Scenario	Description	Requirements
TS-01	Authentication and session management	FR-01, NFR-03
TS-02	Role and permission testing	FR-02, NFR-03
TS-03	Artifact creation and validation	FR-03, FR-15
TS-04	Artifact editing	FR-04
TS-05	Archive/delete	FR-05
TS-06	File upload/retrieval	FR-06, NFR-10
TS-07	Search and filtering	FR-07, FR-08, NFR-01
TS-08	Preservation lifecycle	FR-09
TS-09	Conservation history	FR-10
TS-10	Environmental data and alerts	FR-11, FR-13
TS-11	Reports and dashboard	FR-12
TS-12	Audit logging	FR-14
TS-13	Errors/exceptions	FR-16, NFR-04
TS-14	Performance/security/usability	NFR-01–NFR-08
TS-15	Backup/recovery/availability	NFR-04, NFR-05, NFR-09

6. Test Design Techniques

6.1 Equivalence Partitioning

Inputs are divided into valid and invalid classes. Representative values from each class are tested. For example, supported image/document types form a valid partition while executable or unsupported formats form an invalid partition.

Input	Valid Class	Invalid Class
File type	JPG/PNG/PDF if supported	EXE/script/unsupported
Condition	Configured values	Blank/unsupported
Role	Admin/Curator/Viewer	Unknown/unauthorized
Artifact ID	Unique valid format	Duplicate/malformed/blank
Search	Existing keyword	No-match/invalid input

6.2 Boundary Value Analysis

Values at, immediately below and immediately above configured limits are tested. Exact limits must be taken from the final system specification.

Field	Boundary Tests
Artifact name	Minimum-1, minimum, minimum+1, maximum-1, maximum, maximum+1
File size	Limit-1 byte, exact limit, limit+1 byte
Password	Minimum-1, minimum, minimum+1; maximum if defined
Page size	0, 1, maximum, maximum+1
Humidity	Configured minimum-1, minimum, normal, maximum, maximum+1

6.3 State Transition Testing

A representative artifact lifecycle is Registered → Under Assessment → Stable/Good or Needs Conservation → Under Conservation → Conserved → Archived. Valid transitions should be accepted and invalid transitions rejected.

Current State	Action	Expected State
Registered	Start assessment	Under Assessment
Under Assessment	Assessment complete; good	Stable/Good
Under Assessment	Damage identified	Needs Conservation
Needs Conservation	Start conservation	Under Conservation
Under Conservation	Complete treatment	Conserved
Conserved	Archive approval	Archived
Archived	Normal edit	Reject unless recovery policy permits

7.Exceptional and Negative Conditions

- Database connection lost during create/update.
- Network interruption during file upload.
- Session expires while submitting a form.
- Two users update the same artifact concurrently.
- Two users attempt the same artifact ID simultaneously.

8. Requirement Traceability Matrix

Requirement	Test Cases	Coverage
FR-01	TC-001, TC-002, TC-003, TC-027, NFT-005	Covered
FR-02	TC-004	Covered
FR-03	TC-005–TC-009	Covered
FR-04	TC-010	Covered
FR-05	TC-011, TC-012	Covered
FR-06	TC-013–TC-015, NFT-009	Covered
FR-07	TC-016–TC-018, NFT-001	Covered
FR-08	TC-019	Covered
FR-09	TC-020, TC-021	Covered
FR-10	TC-022	Covered
FR-11	TC-023	Covered
FR-12	TC-024, TC-025	Covered
FR-13	Scenario TS-10; threshold-specific cases depend on final alert specification	Partial
FR-14	TC-026	Covered
FR-15	TC-005, TC-010, TC-028	Covered
FR-16	TC-007, TC-009, TC-014, TC-015, TC-017, TC-021, TC-023, TC-028	Covered
NFR-01–NFR-10	NFT-001–NFT-010	Covered

9. Test Coverage Summary

The suite contains 28 detailed functional test cases and 10 non-functional test cases. It covers normal, invalid, boundary and exceptional conditions, along with security, performance, usability, compatibility, reliability, recovery and data-integrity aspects.

Coverage Type	Included Coverage
Normal	Login, artifact CRUD, upload, search, status, conservation, reports
Boundary	Field length, file size, page size, password and environmental limits
Invalid	Blank data, duplicate IDs, unsupported files, invalid states and values
Exceptional	DB/network failures, session expiry, concurrency and service interruption
Security	RBAC, injection, XSS, session and file-security checks
Reliability	Transactions, backup/recovery and restart
Usability	Forms, navigation, messages and representative workflows
Performance	Search and concurrent-user testing

10. Test Execution and Reporting Guidelines

- Prepare a dedicated test environment and synthetic/anonymized museum data.
- Execute tests using the stated steps and record Actual Result exactly as observed.
- Mark Pass only when actual behavior matches expected behavior; otherwise mark Fail and log a defect ID.
- Re-test fixed defects and execute regression tests for impacted functions.
- Maintain screenshots, logs, API responses and database evidence where appropriate.
- Review the traceability matrix before release to ensure critical requirements are tested.

11. Entry and Exit Criteria

Type	Criteria
Entry	Build deployed; test database/data ready; test accounts available; requirements baselined.
Functional Exit	Critical/high-priority cases executed; no open blocker/critical defects; agreed pass target achieved.
Security Exit	No unresolved critical security findings.
Performance Exit	Defined response-time, throughput and concurrency targets achieved.
Release Exit	Traceability reviewed; regression passed; known defects documented and accepted.

12. Defect Severity

Severity	Meaning	Example
Critical	Core failure, serious data loss or unauthorized access	Artifact data corruption/security breach
High	Major function unavailable or incorrect	Artifact cannot be saved
Medium	Important defect with workaround	Specific report filter incorrect
Low	Minor UI/text issue	Non-critical alignment or wording

13. Conclusion

This test case design provides a structured approach for verifying the Smart Museum Artifact Preservation Platform. Functional and non-functional requirements are mapped to scenarios and detailed cases containing Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result and Pass/Fail Status.

Equivalence Partitioning reduces redundant input testing, Boundary Value Analysis validates limits, Decision Table Testing verifies combinations of conditions, and State Transition Testing verifies the artifact preservation lifecycle. Combined with security, performance, usability, reliability, compatibility and recovery testing, these techniques provide broad requirement and functional coverage.